Vectors I Homework 3: Vector Product — Solutions

Compiled by Maximus Pung Jun

Official worked solutions for sourced past-paper questions; True/False answers from the Basics Worksheet.

1 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{a}\times\mathbf{a}=\mathbf{0}$. $\lvert\mathbf{a}\times\mathbf{a}\rvert=\lvert\mathbf{a}\rvert^2\sin 0^\circ=0$	**T**
(b)	$\mathbf{a}\times\mathbf{a}=0$. $\mathbf{a}\times\mathbf{a}=\mathbf{0}\neq 0$ (left side is the zero vector)	**F**
(c)	$\mathbf{i}\times\mathbf{j}=\mathbf{k}$. $\begin{pmatrix}1\\0\\0\end{pmatrix}\times\begin{pmatrix}0\\1\\0\end{pmatrix}=\begin{pmatrix}0\\0\\1\end{pmatrix}$	**T**
(d)	$\mathbf{a}\times\mathbf{b}=\mathbf{b}\times\mathbf{a}$. $\mathbf{i}\times\mathbf{j}=\mathbf{k},\ \mathbf{j}\times\mathbf{i}=-\mathbf{k}$	**F**
(e)	$\mathbf{a}\times\mathbf{b}=-(\mathbf{b}\times\mathbf{a})$. $\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$	**T**
(f)	$\mathbf{a}\times\mathbf{b}$ is a scalar. $\mathbf{i}\times\mathbf{j}=\mathbf{k}$ (a vector)	**F**
(g)	$\mathbf{k}\times\mathbf{j}=\mathbf{i}$. $\mathbf{j}\times\mathbf{k}=\mathbf{i}\ \therefore\ \mathbf{k}\times\mathbf{j}=-\mathbf{i}$	**F**
(h)	$\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}2\\4\\6\end{pmatrix}=\mathbf{0}$. $\begin{pmatrix}2\\4\\6\end{pmatrix}=2\begin{pmatrix}1\\2\\3\end{pmatrix}$	**T**
(i)	$\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$. Distributive law	**T**
(j)	$(\lambda\mathbf{a})\times\mathbf{b}=\lambda(\mathbf{a}\times\mathbf{b})$ for all $\lambda\in\mathbb{R}$. $(\lambda\mathbf{a})\times\mathbf{b}=\mathbf{a}\times(\lambda\mathbf{b})=\lambda(\mathbf{a}\times\mathbf{b})$	**T**

	Statement	**T/F**
(k)	$\mathbf{a}\times\mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a}=0,\ (\mathbf{a}\times\mathbf{b})\cdot\mathbf{b}=0$	**T**
(l)	$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$ is a scalar. $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\text{vector}\cdot\text{vector}$	**T**
(m)	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. Definition of vector product	**T**
(n)	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. $\lvert\mathbf{i}\times\mathbf{i}\rvert=0\neq 1$	**F**
(o)	If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. $\mathbf{i}\times 2\mathbf{i}=\mathbf{0},\ \mathbf{i}\neq\mathbf{0},\ 2\mathbf{i}\neq\mathbf{0}$	**F**
(p)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ means that $\mathbf{a}$ is parallel to $\mathbf{b}$. $\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta=0\ \therefore\ \theta=0^\circ$ or 180°	**T**
(q)	The area of the parallelogram with adjacent sides $\mathbf{a}$ and $\mathbf{b}$ is $\lvert\mathbf{a}\times\mathbf{b}\rvert$. Area $=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta=\lvert\mathbf{a}\times\mathbf{b}\rvert$	**T**
(r)	The area of triangle OAB is $\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$. Area $=\frac{1}{2}\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$	**F**
(s)	$\mathbf{a}\times(\mathbf{b}\cdot\mathbf{c})$ is a vector. $\mathbf{b}\cdot\mathbf{c}$ is a scalar $\therefore$ meaningless	**F**
(t)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ is perpendicular to $\mathbf{b}$. $\sin\theta=1\ \therefore\ \theta=90^\circ$	**T**
(u)	$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$. both $=\det(\mathbf{a}\ \mathbf{b}\ \mathbf{c})$	**T**
(v)	$\lvert\mathbf{a}\times\mathbf{b}\rvert^2+(\mathbf{a}\cdot\mathbf{b})^2=\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2$. $\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2(\sin^2\theta+\cos^2\theta)=\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2$	**T**
(w)	$\mathbf{a}\times\mathbf{b}=\mathbf{a}\times\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$. $\mathbf{i}\times\mathbf{j}=\mathbf{k}=\mathbf{i}\times(\mathbf{i}+\mathbf{j}),\ \mathbf{j}\neq\mathbf{i}+\mathbf{j}$	**F**
(x)	$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$. $\mathbf{i}\times(\mathbf{i}\times\mathbf{j})=\mathbf{i}\times\mathbf{k}=-\mathbf{j},\ (\mathbf{i}\times\mathbf{i})\times\mathbf{j}=\mathbf{0}$	**F**
(y)	The shortest distance from B to the line through A parallel to $\mathbf{d}$ is $\lvert\overrightarrow{AB}\times\mathbf{d}\rvert$. distance $=\lvert\overrightarrow{AB}\times\hat{\mathbf{d}}\rvert=\dfrac{\lvert\overrightarrow{AB}\times\mathbf{d}\rvert}{\lvert\mathbf{d}\rvert}$	**F**
(z)	If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ and $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. $\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2=\lvert\mathbf{a}\times\mathbf{b}\rvert^2+(\mathbf{a}\cdot\mathbf{b})^2=0$	**T**

2 [JPJC 2024 Promo Q11]

Referred to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-parallel vectors. Point C is the midpoint of OA and point D lies on BC such that $BD:DC=2:3$.

(i) Find the position vectors $\overrightarrow{OC}$ and $\overrightarrow{OD}$, giving your answers in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Find the area of triangle OCD in the form of $k|\mathbf{a}\times\mathbf{b}|$, where k is a constant to be found. [3]

(iii) Given that $|\mathbf{a}|=2$, $|\mathbf{b}|=\sqrt{2}$ and angle AOB is $\frac{\pi}{4}$ radian, using scalar product, show that CD is perpendicular to OA. [3]

(iv) Given instead that $\mathbf{a}$ is a unit vector, give a geometrical interpretation of $|\mathbf{a}\cdot\mathbf{b}|$. [1]

Solution

(i) Given $\overrightarrow{OA}=\mathbf{a}$, $\overrightarrow{OB}=\mathbf{b}$, $\overrightarrow{OC}=\frac{1}{2}\mathbf{a}$.

$$\overrightarrow{OD}=\frac{2\mathbf{c}+3\mathbf{b}}{5}=\frac{1}{5}\left[2\left(\frac{1}{2}\mathbf{a}\right)+3\mathbf{b}\right]=\frac{1}{5}(\mathbf{a}+3\mathbf{b}).$$

(ii)

$$\begin{aligned}\text{Area of triangle } OCD&=\frac{1}{2}|\overrightarrow{OC}\times\overrightarrow{OD}|\\&=\frac{1}{2}\left|\frac{1}{2}\mathbf{a}\times\frac{1}{5}(\mathbf{a}+3\mathbf{b})\right|\\&=\frac{1}{2}\left|\frac{1}{10}(\mathbf{a}\times\mathbf{a})+\frac{3}{10}(\mathbf{a}\times\mathbf{b})\right|\\&=\frac{1}{2}\left|\frac{3}{10}(\mathbf{a}\times\mathbf{b})\right| \qquad (\mathbf{a}\times\mathbf{a}=\mathbf{0})\\&=\frac{3}{20}|\mathbf{a}\times\mathbf{b}|, \qquad k=\frac{3}{20}.\end{aligned}$$

(iii)

$$\overrightarrow{CD}=\overrightarrow{OD}-\overrightarrow{OC}=\frac{1}{5}(\mathbf{a}+3\mathbf{b})-\frac{1}{2}\mathbf{a}=\frac{3}{5}\mathbf{b}-\frac{3}{10}\mathbf{a}$$

$$\begin{aligned}\overrightarrow{CD}\cdot\overrightarrow{OA}&=\left(\frac{3}{5}\mathbf{b}-\frac{3}{10}\mathbf{a}\right)\cdot\mathbf{a}\\&=\frac{3}{5}\mathbf{b}\cdot\mathbf{a}-\frac{3}{10}\mathbf{a}\cdot\mathbf{a}\\&=\frac{3}{5}|\mathbf{b}||\mathbf{a}|\cos\frac{\pi}{4}-\frac{3}{10}|\mathbf{a}|^2\\&=\frac{3}{5}(\sqrt{2})(2)\frac{\sqrt{2}}{2}-\frac{3}{10}(2)^2=\frac{6}{5}-\frac{12}{10}=0.\end{aligned}$$

Since $\overrightarrow{CD}\cdot\overrightarrow{OA}=0$, CD is perpendicular to OA. (Shown)

(iv) $|\mathbf{a}\cdot\mathbf{b}|$ is the length of projection of $\overrightarrow{OB}$ onto $\overrightarrow{OA}$ (or a line or vector parallel to $\overrightarrow{OA}$).

3 [MI 2024 Promo Q4]

Referred to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. C lies on AB such that $AC:AB=1:3$.

(i) Find $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Show that the area of triangle OAC is given by $k|\mathbf{a}\times\mathbf{b}|$, where k is a constant to be determined. [2]

It is now given that $\mathbf{a}$ and $\mathbf{b}$ are unit vectors.

(iii) Using the result in part (i), find the value of $(2\mathbf{a}-\mathbf{b})\cdot\mathbf{c}$. [3]

Solution

(i)

$$AC:AB=1:3 \quad \Rightarrow \quad AC:CB=1:2$$
$$\mathbf{c}=\frac{(1)\mathbf{b}+(2)\mathbf{a}}{1+2}=\frac{2\mathbf{a}+\mathbf{b}}{3}.$$

(ii)

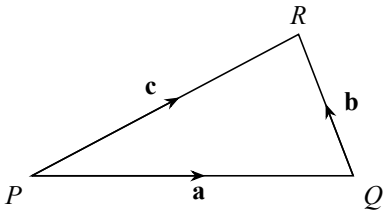
$$\begin{aligned}\text{Area of triangle } OAC &= \frac{1}{3}\text{ Area of triangle } OAB\\ &=\frac{1}{3}\times\frac{1}{2}|\overrightarrow{OA}\times\overrightarrow{OB}|\\ &=\frac{1}{6}|\mathbf{a}\times\mathbf{b}|, \qquad k=\frac{1}{6}.\end{aligned}$$

(iii)

$$\begin{aligned}(2\mathbf{a}-\mathbf{b})\cdot\mathbf{c} &= (2\mathbf{a}-\mathbf{b})\cdot\left(\frac{2\mathbf{a}+\mathbf{b}}{3}\right)\\ &=\frac{1}{3}(2\mathbf{a}-\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})\\ &=\frac{1}{3}(4\mathbf{a}\cdot\mathbf{a}+2\mathbf{a}\cdot\mathbf{b}-2\mathbf{b}\cdot\mathbf{a}-\mathbf{b}\cdot\mathbf{b})\\ &=\frac{1}{3}(4|\mathbf{a}|^2-|\mathbf{b}|^2)\\ &=\frac{1}{3}\left(4(1)^2-(1)^2\right) \quad \text{since } \mathbf{a}, \mathbf{b} \text{ are unit vectors}\\ &=1.\end{aligned}$$

4 [ASRJC 2024 Promo Q5]

In triangle PQR, the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ represent the vectors $\overrightarrow{PQ}$, $\overrightarrow{QR}$ and $\overrightarrow{PR}$ respectively.



(a) State a geometrical interpretation of $|\mathbf{c}\times\hat{\mathbf{a}}|$ and hence show that the area of the triangle PQR can be written as $\frac{1}{2}|\mathbf{c}\times\mathbf{a}|$ units^2. [3]

(b) With clear explanations, show that $|\mathbf{c}\times\mathbf{a}|=|\mathbf{b}\times\mathbf{a}|$ and deduce that

$$\frac{|\mathbf{c}|}{\sin\angle PQR}=\frac{|\mathbf{b}|}{\sin\angle QPR} \quad \text{(Sine Rule).}$$

[2]

Solution

(a) $|\mathbf{c}\times\hat{\mathbf{a}}|$ represents the perpendicular distance from the point R to the line PQ.

$$\begin{aligned}\text{Area of } \triangle PQR &= \frac{1}{2}\times\text{base}\times\text{height}\\ &= \frac{1}{2}|\mathbf{a}|\,|\mathbf{c}\times\hat{\mathbf{a}}|\\ &= \frac{1}{2}|\mathbf{a}|\left|\mathbf{c}\times\frac{\mathbf{a}}{|\mathbf{a}|}\right|\\ &= \frac{1}{2}|\mathbf{c}\times\mathbf{a}|\text{ units}^2. \quad \text{(Shown)}\end{aligned}$$

(b) By replacing $\mathbf{a}$ with $-\mathbf{a}$ and $\mathbf{c}$ with $\mathbf{b}$ in part (a),

$$\begin{aligned}\text{Area of } \triangle PQR &= \frac{1}{2}|\mathbf{b}\times(-\mathbf{a})|=\frac{1}{2}|\mathbf{b}\times\mathbf{a}|\\ \frac{1}{2}|\mathbf{c}\times\mathbf{a}| &= \frac{1}{2}|\mathbf{b}\times\mathbf{a}|\\ |\mathbf{c}\times\mathbf{a}| &= |\mathbf{b}\times\mathbf{a}| \quad \text{(Shown)}\\ |\mathbf{c}||\mathbf{a}|\sin\angle QPR &= |\mathbf{b}||\mathbf{a}|\sin\angle PQR\\ \frac{|\mathbf{c}|}{\sin\angle PQR} &= \frac{|\mathbf{b}|}{\sin\angle QPR}. \quad \text{(Shown)}\end{aligned}$$

5 [CJC 2024 Promo Q7]

(a) Three non-zero vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ are such that $3\mathbf{p}\times\mathbf{q}=\mathbf{r}\times\mathbf{p}$.

Find a linear relationship between $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$. [3]

(b) Referred to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$.

It is given that $\mathbf{a}$ and $\mathbf{b}$ are non-parallel unit vectors.

The point B divides AC in the ratio $1:3$.

The point D lies on OB produced such that $OB:OD=1:m$, where $m\in\mathbb{Z}$, $m\geq 2$.

Given that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{CD}$, find the numerical value of m. [4]

Solution

(a)

$$\begin{aligned}
3\mathbf{p}\times\mathbf{q}&=\mathbf{r}\times\mathbf{p}\\
(3\mathbf{p}\times\mathbf{q})-(\mathbf{r}\times\mathbf{p})&=\mathbf{0}\\
(3\mathbf{p}\times\mathbf{q})+(\mathbf{p}\times\mathbf{r})&=\mathbf{0}\\
\mathbf{p}\times(3\mathbf{q}+\mathbf{r})&=\mathbf{0}\\
\mathbf{p}&\parallel 3\mathbf{q}+\mathbf{r}\\
\mathbf{p}&=k(3\mathbf{q}+\mathbf{r}),\quad k\in\mathbb{R}\setminus\{0\}.
\end{aligned}$$

Verification note (not part of the official scheme): the printed last line assumes $3\mathbf{q}+\mathbf{r}\neq\mathbf{0}$. The question also permits $3\mathbf{q}+\mathbf{r}=\mathbf{0}$. The unrestricted relationship is $3\mathbf{q}+\mathbf{r}=\lambda\mathbf{p}$, $\lambda\in\mathbb{R}$, including $\lambda=0$.

(b)

$$\begin{aligned}
\overrightarrow{OB}&=\frac{3\overrightarrow{OA}+\overrightarrow{OC}}{4}\\
\overrightarrow{OC}&=4\mathbf{b}-3\mathbf{a}\\
\overrightarrow{OD}&=m\overrightarrow{OB}=m\mathbf{b}\\
\overrightarrow{AB}&=\mathbf{b}-\mathbf{a}\\
\overrightarrow{CD}&=m\mathbf{b}-(4\mathbf{b}-3\mathbf{a})=3\mathbf{a}+(m-4)\mathbf{b}.
\end{aligned}$$

Given that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{CD}$,

$$\begin{aligned}
\overrightarrow{AB}\cdot\overrightarrow{CD}&=0\\
(\mathbf{b}-\mathbf{a})\cdot[3\mathbf{a}+(m-4)\mathbf{b}]&=0\\
3\mathbf{a}\cdot\mathbf{b}+(m-4)\mathbf{b}\cdot\mathbf{b}-3\mathbf{a}\cdot\mathbf{a}-(m-4)\mathbf{a}\cdot\mathbf{b}&=0\\
3\mathbf{a}\cdot\mathbf{b}+(m-4)|\mathbf{b}|^2-3|\mathbf{a}|^2-(m-4)\mathbf{a}\cdot\mathbf{b}&=0\\
(m-4)-3-(m-7)\mathbf{a}\cdot\mathbf{b}&=0\\
m(1-\mathbf{a}\cdot\mathbf{b})&=3-7\mathbf{a}\cdot\mathbf{b}+4\\
m&=\frac{3-7\mathbf{a}\cdot\mathbf{b}+4}{1-\mathbf{a}\cdot\mathbf{b}}=\frac{7(1-\mathbf{a}\cdot\mathbf{b})}{1-\mathbf{a}\cdot\mathbf{b}}=7.
\end{aligned}$$

6 [SAJC 2024 Promo Q9]

(a) Relative to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel. It is given that $|\mathbf{a}|=9$, $|\mathbf{b}|=1$ and $|2\mathbf{a}+\mathbf{b}|=2\sqrt{74}$.

(i) By considering $(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=-\frac{29}{4}$. [3]

(ii) Give a geometrical meaning of $|\mathbf{a}\cdot\mathbf{b}|$. [1]

(iii) The points P, Q and R have position vectors $7\mathbf{a}-5\mathbf{b}$, $6\mathbf{a}+5\mathbf{b}$ and $9\mathbf{a}+\lambda\mathbf{b}$ respectively. Given that these three points are collinear, find the value of λ. [3]

(b) In the triangle OUV where O is the origin, the position vectors of the points U and V are $\mathbf{u}$ and $\mathbf{v}$ respectively. The point W is the midpoint of OU, and the point X has position vector given by $\frac{1}{16}\mathbf{u}+\frac{3}{4}\mathbf{v}$. Show that the area of triangle UWX can be written as $m|\mathbf{u}\times\mathbf{v}|$ where m is a constant to be found. [3]

Solution

(a)(i)

$$\begin{aligned}(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b}) &= 4(\mathbf{a}\cdot\mathbf{a})+2(\mathbf{b}\cdot\mathbf{a})+2(\mathbf{a}\cdot\mathbf{b})+(\mathbf{b}\cdot\mathbf{b})\\ &= 4|\mathbf{a}|^2+4(\mathbf{a}\cdot\mathbf{b})+|\mathbf{b}|^2\\ |2\mathbf{a}+\mathbf{b}|^2 &= 4|\mathbf{a}|^2+4(\mathbf{a}\cdot\mathbf{b})+|\mathbf{b}|^2\\ (2\sqrt{74})^2 &= 4(9^2)+4(\mathbf{a}\cdot\mathbf{b})+(1^2)\\ 296 &= 4(9^2)+4(\mathbf{a}\cdot\mathbf{b})+(1^2)\\ 4(\mathbf{a}\cdot\mathbf{b}) &= 296-4(9^2)-(1^2)\\ \mathbf{a}\cdot\mathbf{b} &= -\frac{29}{4}.\end{aligned}$$

(a)(ii) $|\mathbf{a}\cdot\mathbf{b}|$ is the length of projection of $\overrightarrow{OA}$ onto $\overrightarrow{OB}$.

(a)(iii) If P, Q and R are collinear, then $k\overrightarrow{PQ}=\overrightarrow{PR}$.

$$\begin{aligned}k[(6\mathbf{a}+5\mathbf{b})-(7\mathbf{a}-5\mathbf{b})] &= [(9\mathbf{a}+\lambda\mathbf{b})-(7\mathbf{a}-5\mathbf{b})]\\ k[-\mathbf{a}+10\mathbf{b}] &= [2\mathbf{a}+(\lambda+5)\mathbf{b}].\end{aligned}$$

Since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel, non-zero vectors,

$$\begin{aligned}k &= -2\\ \lambda+5 &= 10k=-20\\ \lambda &= -25.\end{aligned}$$

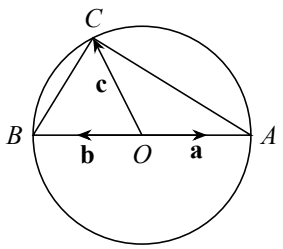
(b)

$$\begin{aligned}\overrightarrow{UX} &= \overrightarrow{OX}-\overrightarrow{OU}=\frac{1}{16}\mathbf{u}+\frac{3}{4}\mathbf{v}-\mathbf{u}=\frac{3}{4}\mathbf{v}-\frac{15}{16}\mathbf{u}\\ \overrightarrow{UW} &= \overrightarrow{OW}-\overrightarrow{OU}=\frac{1}{2}\mathbf{u}-\mathbf{u}=-\frac{1}{2}\mathbf{u}\end{aligned}$$

$$\begin{aligned}\text{Area of triangle } UWX &= \frac{1}{2}|\overrightarrow{UW}\times\overrightarrow{UX}|\\ &= \frac{1}{2}\left|\frac{1}{2}\mathbf{u}\times\left(\frac{3}{4}\mathbf{v}-\frac{15}{16}\mathbf{u}\right)\right|\\ &= \frac{3}{16}|\mathbf{u}\times\mathbf{v}| \qquad (\mathbf{u}\times\mathbf{u}=\mathbf{0})\\ m &= \frac{3}{16}.\end{aligned}$$

7 [CJC 2025 Promo Q6]

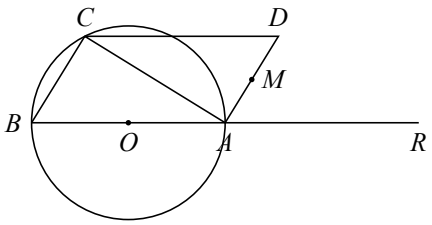
The diagram shows a unit circle with centre O and diameter AB. The point C lies on the circumference of the circle. The position vectors of A, B and C are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively.



(a) By considering an appropriate scalar product, show that AC and BC are perpendicular. [4]

(b) Give a geometrical interpretation of $|\mathbf{a}\cdot(\mathbf{c}-\mathbf{a})|$. [1]

The point D is such that $ABCD$ is a parallelogram. The point M is the mid-point of AD and the point R lies on OA produced such that $OA:AR=1:2$.



(c) By considering $\overrightarrow{CM}\times\overrightarrow{CR}$ or otherwise, show that the points C, M and R are collinear. [3]

Solution

(a)

$$\begin{aligned}
\overrightarrow{AC}&=\mathbf{c}-\mathbf{a}, \qquad \overrightarrow{BC}=\mathbf{c}-\mathbf{b}\\
\overrightarrow{AC}\cdot\overrightarrow{BC}&=(\mathbf{c}-\mathbf{a})\cdot(\mathbf{c}-\mathbf{b})\\
&=(\mathbf{c}-\mathbf{a})\cdot(\mathbf{c}-(-\mathbf{a}))\\
&=(\mathbf{c}-\mathbf{a})\cdot(\mathbf{c}+\mathbf{a})\\
&=\mathbf{c}\cdot\mathbf{c}-\mathbf{a}\cdot\mathbf{a}\\
&=|\mathbf{c}|^2-|\mathbf{a}|^2\\
&=0 \qquad (|\mathbf{c}|=|\mathbf{a}|=\text{radius of circle}).
\end{aligned}$$

Hence AC and BC are perpendicular.

(b) $|\mathbf{a}\cdot(\mathbf{c}-\mathbf{a})|$ is the length of projection of $\overrightarrow{AC}$ onto $\overrightarrow{OA}$.

(c) By Ratio Theorem,

$$\begin{aligned}
\overrightarrow{OM} &= \frac{\overrightarrow{OD}+\overrightarrow{OA}}{2} \\
&= \frac{1}{2}(\mathbf{c}+2\mathbf{a}+\mathbf{a}) = \frac{1}{2}\mathbf{c}+\frac{3}{2}\mathbf{a} \\
\overrightarrow{CM}\times\overrightarrow{CR} &= (\overrightarrow{OM}-\overrightarrow{OC})\times(\overrightarrow{OR}-\overrightarrow{OC}) \\
&= \left[\frac{1}{2}\mathbf{c}+\frac{3}{2}\mathbf{a}-\mathbf{c}\right]\times(3\mathbf{a}-\mathbf{c}) \\
&= \left(\frac{3}{2}\mathbf{a}-\frac{1}{2}\mathbf{c}\right)\times(3\mathbf{a}-\mathbf{c}) \\
&= \frac{9}{2}\mathbf{a}\times\mathbf{a}-\frac{3}{2}\mathbf{a}\times\mathbf{c}-\frac{3}{2}\mathbf{c}\times\mathbf{a}+\frac{1}{2}\mathbf{c}\times\mathbf{c} \\
&= \mathbf{0}-\frac{3}{2}\mathbf{a}\times\mathbf{c}+\frac{3}{2}\mathbf{a}\times\mathbf{c}+\mathbf{0} \\
&= \mathbf{0}.
\end{aligned}$$

$\overrightarrow{CM}$ and $\overrightarrow{CR}$ are parallel. Since CR and CM are parallel and C is a common point, hence C, M and R are collinear.

END OF HOMEWORK